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Scheduling trucks and drones for cooperative deliveries

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ABSTRACT

Truck and drone based cooperative delivery system is an emerging instant delivery mode for transporting packages in a more timely and efficient way than traditional delivery mode. This paper studies the scheduling of a fleet of truck groups, each of which can carry multiple drones. We formulate a mixed integer programming model for truck groups routes, and for the timing of the drones' launching and return to their dedicated truck. The model objective is to minimize the total operational cost, including truck travel cost, drone travel cost, using truck groups fixed cost, and potential penalty for late delivery at customer locations. Column generation-based heuristic algorithm and some acceleration techniques are designed for solving the model efficiently. We carry out numerical experiments for validating the effectiveness of the mathematical model, algorithm, and acceleration techniques. In addition, sensitivity analyses are performed to derive some managerial implications.

1. Introduction

During the times of social distancing due to the SARS-COV-2 pandemic, drones became a common feature in the express delivery industry. In 2022, forced isolation in some Chinese metropolises has led to an increase in food, medicine and package delivery systems. A number of Chinese hospitals and centers of disease control and prevention have used drones to deliver medical samples and medicines in a timely way. Several delivery companies have also conducted tests for the use of drones in food delivery. Shifting from traditional urban traffic to urban air traffic will free up air traffic capacity. Wang and Qu (2023) explored the future development of urban aerial mobility, which includes the possibility of using drones and others for cargo distribution. The Civil Aviation Administration of China has issued a letter of approval for the trail operation of specific types of drones and licenses for the use of delivery drones by some logistics companies. It is anticipated that drone-based delivery will become more prevalent in the coming years.

UPS company conducted a drone delivery trial in Florida in 2017. The driver places the cargo on the bottom of the drone and releases it through the cockpit control panel while he drives to the next destination. UPS estimates that the use of drones saves $50 million a year by reducing the last mile of delivery (MH, & Staff, L., 2017). While drones have the advantages of overcoming traffic limitations in urban areas and limitations posed by the landscape in rural areas, their coverage radius is constrained by their cruising power, which implies that cooperation between trucks and drones is desirable. In addition, Qu et al. (2022) proposed a mode in which drones and passenger vehicles work together to deliver parcels. In this mode, the drone picks up a parcel from a depot and drops it onto

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the roof-rack of a passenger vehicle. The passenger vehicle will deliver the parcel to the destination area, and then another drone will fly to the passenger vehicle to pick up the parcel and deliver it to the customer. This mode makes full use of the roof resources of passenger vehicles and can replace traditional truck transportation. Similar to the above mode, this paper studies a truck-drone cooperative mode. A fleet of truck groups carrying drones delivers goods to customers. A truck group consists of one truck and multiple drones, in which these drones are loaded on the truck. The delivered packages are also carried by the truck. Under this truck-drone cooperative mode, the coverage radius can be extended significantly.

In order to operate this cooperative delivery mode, one of the core decisions is how to schedule trucks and drones to efficiently satisfy the customers' delivery requirements. This study proposes a comprehensive decision model for route scheduling of multiple truck groups for package delivery. The decision includes truck routes, drone routes, and the time at which and locations where drones are launched from and returned to their dedicated trucks. The model aims to minimize the total operational cost, including truck travel cost, drone travel cost, using truck groups fixed cost, and potential penalty for late delivery at customer locations; while most of related literature only minimizes the makespan. In addition, our model is formulated in the context of a road network containing more nodes than in previous related studies. In order to solve this model, a novel solution method that is column generation-based heuristic algorithm is developed; while most of related literature adopts the neighborhood search based algorithmic framework. Some new acceleration techniques are also designed and implemented to shorten the solution time. Based on a company's operation network of drones in Hangzhou city, extensive experiments are conducted to validate the components performance, these components are embedded in column generation-based heuristic algorithm. Managerial insights are also provided through several sensitivity analyses. Our methodology is applicable to other cooperative systems using trucks or vans, and sidekicks such as autonomous robots. The contributions of this study as follows.

1) We establish a comprehensive decision model for route scheduling of multiple truck groups for package delivery, with the goal to minimize the total operational cost, including truck travel cost, drone travel cost, using truck groups fixed cost, and potential penalty for late delivery at customer locations.

2) A novel solution method that is column generation-based heuristic algorithm is developed to solve this complex problem. In the process of algorithm development, we designed a variable neighborhood search algorithm to solve the pricing problem.

3) Many numerical experiments are carried out based on a company's operation network of drones in Hangzhou city. The results show that the algorithm has good performance in terms of solution quality and computation time. In addition, the proposed algorithm can work in a rolling horizon manner and solve a batch of orders containing about 30 orders every 15 min. In other words, in a realistic context, in which orders emerge and enter the system at an average rate of 120 orders per hour, our proposed methodology can handle the accumulated orders every 15 min.

The paper's remainder is organized as follows. Section 2 The related works are reviewed in Section 2. Section 3 describes the problem background. A mixed integer linear programming (MILP) model is formulated in Section 4, and Section 5 designs a column generation-based heuristic algorithm. Section 6 reports the results of numerical experiments. The conclusions are outlined in Section 7.

2. Related works

Trucks and drones cooperative delivery problems have been extensively studied and discussed in the research community, and several reviews have been written (see Chung et al., 2020; Coutinho et al., 2018; Khoufi et al., 2019; Liang & Luo, 2022; Macrina et al., 2020; Otto et al., 2018; Rojas Viloria et al., 2021). Otto et al. (2018) investigated and analyzed the most promising applications in the field of civil applications of drones, focusing on optimization methods for civil applications of drones. Their review covered more than 200 references. Coutinho et al. (2018) defined drone routing and trajectory optimization problems. They introduced a taxonomy applied to 70 papers. At the same time, the latest contributions of drone trajectory optimization and drone path selection were surveyed. Khoufi et al. (2019) reviewed the drone path optimization problem and introduced a classification method, specifically considering the vehicles type and number, the routing problem type, solution techniques, and applications. Regarding trucks assisted by one or several drones, they reviewed papers from three aspects: delivery by a truck and a drone, delivery by a truck and multiple drones, and delivery by multiple trucks and multiple drones. Chung et al. (2020) investigated problems arising in drones and drone-truck joint operations and introduced some application fields. Meanwhile, Macrina et al. (2020) reviewed 63 articles focusing on the problem of delivery routing. They surveyed existing works and provided perspectives for future research. Unlike these survey papers, we consider the case where each customer is served by a drone (never by a truck). In addition, Rojas Viloria et al. (2021) analyzed papers on drone routing between 2005 and 2019. Liang & Luo (2022) comprehensively reviewed the routing problem from the perspective of models and algorithms. They summarized the main characteristics considered in this problem.

In this paper, the trucks can carry multiple drones. According to Khoufi et al. (2019) for their work in delivery transportation, we extend studies on truck-drone cooperative systems. This section reviews the related works under four headings: the traveling salesman problem with drones (TSP-D) with one drone, the TSP-D with multiple drones, the traveling salesman problem with drones (VRP-D) with one drone on one truck, and the VRP-D with multiple drones on one truck.

2.1. The TSP-D with one drone

Murray & Chu (2015) first studied the TSP-D with one drone on one truck. They proposed two kinds of delivery problems between trucks and drones. Both types of problems assume a depot and some customers, each of which is served by only one truck or one drone, and the optimization goal is minimizing the latest delivery time. to complete delivery tasks. Carlsson & Song (2018) studied this problem and argued that the truck and the drone should be allowed to meet anywhere, not just at a depot or at customer locations.

Agatz et al. (2018) developed a mathematical model and proposed a two-stage heuristic method. Ha et al. (2018) studied a new variety of TSP-D, presented a mathematical model, and designed two local search heuristics for this problem. Ha et al. (2020) extended the study by Ha et al. (2018) and presented a new hybrid genetic algorithm. Several exact algorithms have been designed for the TSP-D with one drone on one truck. Thus Dell'Amico et al. (2021) proposed and developed a branch-and-cut algorithms for solving TSP-D. Boccia et al. (2021) established a mathematical model and proposed a column-and-row generation approach. There are several additional papers on this problem, such as those of El-Adle et al. (2021), Roberti & Ruthmair (2021), Yurek & Ozmutlu (2021), and Dell'Amico et al. (2022). In addition, from the perspective of modeling, the van and robot cooperative delivery problem is similar to the truck and drones cooperative delivery problem. For example, Yu et al. (2022) assumed that each van carries one robot. They studied the pickup and delivery routing problem based on a van-robot, and designed a heuristic algorithm.

2.2. The TSP-D with multiple drones

Phan et al. (2018) investigated an extension of the TSP-D with multiple drones, then extended the algorithm proposed by Ha et al. (2018). Karak & Abdelghany (2019) presented a TSP-D with multiple drones for pickup and delivery services, which divides nodes into a depot, customers, and stations. In addition, all customers were served by drones and the truck only as a mobile depot. They provided a model and designed a extending the classical Clarke and Wright (1964) algorithm to solve it. Moshref-Javadi et al. (2020) considered a system based on multiple drones and a truck, presented a mathematical model for minimizing customers' waiting time, and proposed a routing heuristic algorithm. Moshref-Javadi et al. (2021) conducted a comparative evaluation of three synchronous truck-drone models for last-mile delivery. The three delivery models consist of a truck and multiple drones. Murray & Raj (2020) considered the fact that the flight time for drones is affected by battery size, flight distance, payload, and other factors. They presented a mathematical model of the TSP-D with multiple drones. This kind of problem considered using a set of different types of drones to work in tandem with a truck to provide delivery services to customers. On the basis of the work of Murray & Raj (2020), Raj & Murray (2020) considered varying speeds of drones, and provided a heuristic method. Luo et al. (2021a, 2021b) considered the energy consumption, and decomposed the TSP-D with multiple drones into three subproblems, established an MILP model with makespan minimization, and proposed two algorithms. Luo et al. (2021a, 2021b) considered flexible time windows, and designed a multi-objective optimization approach to solve a routing problem with multiple drones. Kang & Lee (2021) considered drone speed and battery capacity to study the TSP-D with multiple drones on one truck, and designed an exact algorithm. Leon-Blanco et al. (2022) allowed drones to serve multiple customers per trip, and designed an agent-based method to handle large instances. In addition, Lin et al. (2020) studied a multi-drone single-truck distribution system in which drones serve all customers.

2.3. The VRP-D with one drone on one truck

Different from the TSP-D, the VRP-D implies there are multiple truck groups. In addition, the publications on the TSP-D (reviewed in Sections 2.1 and 2.2) usually do not consider the truckload capacity. We first review the VRP-D related works, where each truck carries a drone. Sacramento et al. (2019) proposed a VRP-D delivery model and developed an algorithm for solving this model. The problem studied by Sacramento et al. (2019) considered that drones could take off and land at customer nodes along truck routes, so that trucks and drones could provide delivery services to customers in parallel. According to the assumption of Sacramento et al. (2019), Zhen et al. (2023) proposed an MILP model that further considered that the drone flying duration is affected by customer demands, and designed a branch-price-and-cut algorithm. Kuo et al. (2022) established an MILP model given time windows, which extended the model by Sacramento et al. (2019), and proposed an algorithm for this MILP model. Gu et al. (2022) and Lei et al. (2022) studied the VRP-D with a truck carrying a drone.

2.4. The VRP-D with multiple drones on one truck

Based on the VRP-D with one drone on one truck, some scholars have studied the VRP-D with multiple drones on one truck, which is the case most similar to that of this study. Wang et al. (2017) investigated this problem from a worst-case perspective. Poikonen et al. (2017) studied the VRP-D. In this problem, there was some vehicles and a certain number of drones, trucks and drones working together to perform delivery tasks. Schermer et al. (2019) studied the VRP-D with multiple drones on one truck. These authors proposed a mathematical model and designed a metaheuristic. Wang & Sheu (2019) studied a VRP-D with restrictions on synchronization locations. The problem had strict requirements for drone landing, that was, the drone must land at the hub station. In this way, not only could the truck serve the customer, but the drone on board could also leave the truck to serve one or more customers, and then the truck could pick up the drone at the hub for the next delivery. Kitjacharoenchai et al. (2020) introduced a vehicle routing model considering multiple drones on a truck, and designed two heuristics including a large neighborhood search and a truck and drone route construction procedure. Li te al. (2020) studied the VRP-D with multiple drones and proposed a heuristic method. Tamke & Buscher (2021) established a mathematical model and proposed an exact algorithm. Considering the developing technology that drones can be launched or landed from moving trucks without stopping, Li et al. (2022) also developed an MILP model and designed a heuristic method. Lin et al. (2022) studied a multi-truck transportation system in which there were multiple drones on a truck. They considered the routing problem of medical resource delivery system, and designed a hybrid algorithm for it. Yan et al. (2022) considered time windows and random attacks to optimize truck-drone routes executing reconnaissance tasks. They proposed an algorithm. Similar to VRP-D with multiple drones on one truck, Chen et al. (2021) considered time windows and proposed a van routing problem with multiple delivery robots on one van and developed a heuristic method.

Table 1 lists the main characteristics considered in these papers presented in Sections 2.2 and 2.4, in which each truck carries multiple drones. In the reviewed literature, Karak & Abdelghany (2019), Lin et al. (2020), and Lin et al. (2022) considered nodes other than depot and customers. In addition, Luo et al. (2021a, 2021b), Li et al. (2020), Li et al. (2022), Yan et al. (2022), and Chen et al. (2021) included time windows. The above two features are also considered in this paper. In the existing literature, most of the nodes involved in the literature are only depot and customer nodes, and the maximum number of customers that can be resolved within a limited time range from 50 to 1000. Unlike the existing literature, our model considers customer nodes, depot, and road network nodes, and each road network node is not necessarily visited, which makes our model more complex. In addition, our proposed algorithm can work in a rolling horizon manner, processing 30 orders every 15 min. In a real-world environment, our truck-drone delivery system can handle 120 orders per hour.

3. Problem description

This paper studies the cooperative delivery of the parcel delivery tasks by multiple trucks groups. A truck group consists of one truck and multiple drones, in which these drones are loaded on the truck. For each truck group, the truck carries drones, and travels on a road network. The drones can detach from the truck at a road network node, deliver the goods to a customer, and return to the truck at a road network node. During the drones' flights for deliveries, the trucks need not stop to wait for the drones, but continue to drive. In this problem, the trucks can only travel between nodes in the road network. Each customer is served by drones, and only once.

The problem studied can be described on the graph $G = (N, A)$. all nodes set on the plane is $N = C \cup O$, where C is all customers set, which need to be delivered, O is all road network nodes set. All arcs set in the plane is $A = A^K \cup A^D = \{(i,j)|i,j \in N\}$, where A^K is all truck arcs set, and A^D is all drone arcs set. $|K|$ truck groups start from a depot, and each truck carries $|D|$ drones. Drone d on truck k must take off from it and return to it after serving customers. The maximum weight of the goods that can be transported by a truck is m^K, and the quantity demanded by customer i is q_i. Parameter t_{ij}^k represents the truck driving time between node i and node j, t_{ij}^D is the drone flying time between node i and node j. The latest time for drones arriving at customer c is denoted by e_c; a cost penalty equal to s^T per time unit is incurred, when the arrival time at customer c is later than e_c. In addition, s^K is the truck travel cost per unit time, s^D are the drone travel cost per unit time, and s^G is the per using truck group fixed cost.

The problem considered in this study is to schedule all truck group routes so as minimizing the total operational cost, including truck travel cost, drone travel cost, using truck groups fixed cost, and potential penalty for late delivery at customer locations.

Before we establish the model, we simplify the problem by assuming conditions to facilitate modeling. We summarize its basic assumptions as follows.

Table 1
The main characteristics for problems of each truck carrying multiple drones.

Authors and years	Nodes other than depot and customers	Number of trucks	Objective		Time window of customer	Maximum number of customers	Algorithm
			time	cost			
Phan et al. (2018)		1		✓		100	ALNS
Karak & Abdelghany (2019)	✓	1		✓		100	HCWH
Moshref-Javadi et al. (2020)		1	✓			101	TDRA
Murray & Raj (2020)		1	✓			100	Three-phase heuristic
Raj & Murray (2020)		1	✓			100	Three-phase heuristic
Lin et al.(2020)	✓	1	✓			100	GA and PSO
Luo et al. (2021a, 2021b)		1	✓			100	MSTS
Luo et al. (2021a, 2021b)		1		✓	✓	80	HMOA
Kang & Lee (2021)		1	✓			50	Logic-based Benders decomposition
Leon-Blanco et al. (2022)		1	✓			500	Agent-based approach
Moshref-Javadi et al. (2021)		1	✓			1000	TDRA
Poikonen et al. (2017)		n	✓			–	Worst-case analysis
Wang et al. (2017)		n	✓			–	Worst-case analysis
Schermer et al. (2019)		n	✓			100	Matheuristic
Kitjacharoenchai et al. (2020)		n	✓			80	DTRC and LNS
Li et al. (2020)		n		✓	✓	100	ALNS
Tamke & Buscher (2021)		n	✓	✓		100	Branch-and-cut
Li et al. (2022)		n		✓	✓	100	ALNS
Lin et al. (2022)	✓	n	✓			100	PSO, GA and an approximation method
Yan et al. (2022)		n		✓	✓	100	ALNS-SP
Chen et al. (2021)		n	✓		✓	200	ALNS
This paper	✓	n		✓	✓	120	CG

Notes: The Maximum number of customers column represents the number of customers that can be served in an hour. ALNS: adaptive large neighborhood search; HCWH: hybrid Clarke and Wright heuristic; TDRA: truck and drone routing algorithm; MSTS: multi-start tabu search; HMOA: hybrid multi-objective optimization approach; PSO: particle swarm optimization; GA: genetic algorithm; Matheuristics: embed mathematical programming inside a heuristic framework; ALNS-SP: Adaptive Large Neighborhood Search with Short-path; CG: Column generation. The delivery robot in Chen et al. (2021) is similar to the drone in other papers mentioned.

(1) A truck carries multiple drones, which can only be launched and returned from dedicated truck at a road network node or the depot.

(2) A drone visits one customer per flying trip.

(3) Assume that each truck has enough batteries, and we replace the batteries time is short, so the time to replace the batteries can be ignored in the mathematical model. In addition, we assume that for each customer, there are at least two arcs a_1 and a_2, such that the time it takes for the drone to fly between a_1 and a_2 is no higher than the maximum flight time of the drone, $a_1,\ a_2 \in A^D$.

4. Mathematical formulation

In order to optimize the truck group routes, we develop an MILP model in this section. Before giving the mathematical formulation, we describe the used notation.

4.1. Notation

Indices and sets

D the set of all drones on each truck, which is indexed by d;
K the set of all truck groups, which is indexed by k;
O the set of all road network nodes, which is indexed by i,j; $O = \{0,1,2,\cdots,o,o+1\}$; 0 and $o+1$ represent the same depot;
C the set of all customers, $C = \{o+2, o+3, \cdots, o+1+c\}$, which is indexed by c,i,j;
N the set of all nodes, $N = C \cup O$, which is indexed by i,j,c.
O_- the set of all road network nodes from which trucks or drones may depart, $O_- = O\backslash\{o+1\} = \{0,1,2,\cdots,o\}$;
O_+ the set of all road network nodes at which trucks or drones may arrive, $O_+ = O\backslash\{0\} = \{1,2,\cdots,o,o+1\}$;
O_0 the set of all road network nodes without depots, $O_0 = O\backslash\{0,o+1\} = \{1,2,\cdots,o\}$;

Parameters

t_{ij}^K truck driving time between node i and node j;
t_{ij}^D drone flying time between node i and node j;
q_i quantity demanded by customer i;
m^K maximum load capacity of truck;
M a sufficiently positive number;
e_c the latest time for drones arriving at customer c;
s^K the truck travel cost per unit time;
s^D the drone travel cost per unit time;
s^T the potential penalty per unit time;
s^G the per using truck group fixed cost.

Decision variables

ε_k binary, if truck group k is used, $\varepsilon_k = 1$;
α_{ijk} binary, if truck k travels from node $i \in O_-$ to node $j \in O_+$, $\alpha_{ijk} = 1$;
β_{ijdk} binary, if drone d on truck k flies from node $i \in N$, then flies to node $j \in N$, $\beta_{ijdk} = 1$;
μ_{ik} integer, if truck k starts from the depot, the μ_{ik} th node to be visited is node i;
φ_{idk} integer, if drone d on truck k starts from the depot, the φ_{idk} th node to be visited is node i;
δ_{ik} continuous nonnegative, the latest time for truck k and drones of truck k at node i.

4.2. Mathematical model

We formulate the following MILP model:

$$[\text{M1}]\ \text{Minimize}\left\{\sum_{k\in K}s^G\varepsilon_k+\sum_{k\in K}\sum_{i\in O_-}\sum_{j\in O_+}s^Kt_{ij}^K\alpha_{ijk}+\sum_{k\in K}\sum_{d\in D}\sum_{i\in O_-}\sum_{c\in C}s^Dt_{ic}^D\beta_{icdk}+\sum_{k\in K}\sum_{d\in D}\sum_{j\in O_+}\sum_{c\in C}s^Dt_{jc}^D\beta_{cjdk}+\sum_{k\in K}\sum_{c\in C}s^T\left(\delta_{ck}-e_c\sum_{d\in D}\sum_{i\in O_-}\beta_{icdk}\right)^+\right\} \tag{1}$$

subject to

$$\sum_{j\in O_+}\alpha_{0jk}=1 \quad k\in K \tag{2}$$

$$\sum_{i\in O_-}\alpha_{i(o+1)k}=1 \quad k\in K \tag{3}$$

$$\sum_{i\in O_-}\alpha_{ijk}=\sum_{i\in O_+}\alpha_{jik}\leq 1\ j\in O_0,k\in K \tag{4}$$

$$\sum_{j\in O_+\cup C}\beta_{0jdk}=1 \quad k\in K,d\in D \tag{5}$$

$$\sum_{i\in O_-\cup C}\beta_{i(o+1)dk}=1 \quad k\in K,d\in D \tag{6}$$

$$\sum_{i\in O_-\cup C}\beta_{ijdk}=\sum_{i\in O_+\cup C}\beta_{jidk}\leq 1 j\in O_0\cup C,k\in K,d\in D \tag{7}$$

$$\sum_{k\in K}\sum_{i\in O_-}\sum_{d\in D}\beta_{icdk}=1 \quad c\in C \tag{8}$$

$$\sum_{k\in K}\sum_{j\in O_+}\sum_{d\in D}\beta_{cjdk}=1 \quad c\in C \tag{9}$$

$$\beta_{ijdk}=0 \quad i\in C,j\in C,k\in K,d\in D \tag{10}$$

$$\sum_{c\in C}\beta_{icdk}\leq 1 \quad i\in O_-,k\in K,d\in D \tag{11}$$

$$\sum_{c\in C}\beta_{cjdk}\leq 1 \quad j\in O_+,k\in K,d\in D \tag{12}$$

$$\delta_{0k}=0 \quad k\in K \tag{13}$$

$$\delta_{jk}\geq\delta_{ik}+t_{ij}^{K}-M\left(1-\alpha_{ijk}\right) i\in O_-,j\in O_+,k\in K \tag{14}$$

$$\delta_{ck}\geq\delta_{ik}+t_{ic}^{D}-M(1-\beta_{icdk}) i\in O_-,c\in C,k\in K,d\in D \tag{15}$$

$$\delta_{jk}\geq\delta_{ck}+t_{jc}^{D}-M\left(1-\beta_{cjdk}\right) c\in C,j\in O_+,k\in K,d\in D \tag{16}$$

$$\delta_{jk}\geq\delta_{ik}+t_{ij}^{K}-M\left(1-\beta_{ijdk}\right) i\in O_-,j\in O_+,d\in D,k\in K \tag{17}$$

$$\delta_{jk}\geq\delta_{ik}-M\left(2-\beta_{icdk}-\beta_{cjdk}\right) c\in C,i\in O_-,j\in O_+,k\in K,d\in D \tag{18}$$

$$\beta_{ijdk}\leq\alpha_{ijk} \quad i\in O_-,j\in O_+,k\in K,d\in D \tag{19}$$

$$\beta_{icdk}\leq\sum_{j\in O_+}\alpha_{ijk} \quad c\in C,i\in O_-,k\in K,d\in D \tag{20}$$

$$\beta_{cjdk}\leq\sum_{i\in O_-}\alpha_{ijk} \quad c\in C,j\in O_+,k\in K,d\in D \tag{21}$$

$$\sum_{i\in O_-}\sum_{c\in C}\sum_{d\in D}q_c\beta_{icdk}\leq m^K k\in K \tag{22}$$

$$\alpha_{0(o+1)k}=1-\varepsilon_k \quad k\in K \tag{23}$$

$$\varepsilon_k\geq\sum_{i\in O_-}\beta_{icdk} \quad c\in C,k\in K,d\in D \tag{24}$$

$$\varphi_{idk}-\varphi_{jdk}\leq(|O|+|C|)\left(1-\beta_{ijdk}\right)-1 i\in O_-\cup C,j\in O_+\cup C,k\in K,d\in D \tag{25}$$

$$\mu_{ik}-\mu_{jk}\leq|O|\left(1-\alpha_{ijk}\right)-1 i\in O_-,j\in O_+,k\in K \tag{26}$$

$$\delta_{ik}\geq 0 \quad i\in N,k\in K \tag{27}$$

$$0\leq\varphi_{idk}\leq|O|+|C| \quad i\in N,k\in K,d\in D \tag{28}$$

$$0\leq\mu_{ik}\leq|O| \quad i\in N,k\in K \tag{29}$$

$$\alpha_{ijk},\beta_{ijdk},\varepsilon_k\in\{0,1\} i,j\in N,k\in K,d\in D \tag{30}$$

Objective (1) minimizes the total operational cost, including truck travel cost, drone travel cost, using truck groups fixed cost, and potential penalty for late delivery at customer locations. Constraints (2) and (3) indicate that truck k initially leaves the depot and ends up back at the depot. Constraints (4) enforce flow conservation. Each drone initially leaves the depot and ends up back at the depot after completing all distribution tasks in Constraints (5) and (6). Constraints (7) are another family of flow conservation constraints. Constraints (8) and (9) mean that customers can only be served by drones, and one customer can only be served by one drone. In Constraints (10), each drone can serve one customer immediately after leaving the truck. Constraints (11) and (12) indicate that drone d on truck k leaves the truck at most once at node i and returns to the truck for at most once at node j. Constraints (13) state that the time at which truck group k depart from the depot is zero. Constraints (14) calculate trucks travel time between two nodes. Constraints (15) require that the time for truck group k to leave customer c should not be less than the time for truck group k to leave node i, plus drone travel time from node i to customer c. Constraints (16) state truck group k time to leave node j is not less than the time for truck group k to leave customer c, plus drone travel time from customer c to node j. Constraints (17) indicate that if truck k and drone d on truck k starts from node i and arrives at node j, then the time for truck group k to leave node j should be no earlier than the time for truck group k to leave node i, plus truck k travel time from node i to node j. Constraints (18) mean that if drone d on truck k arrives at node j after starting from node i to serve customer c, the time for truck group k to leave node j should be later than the time for truck group k to leave node i. Constraints (19)–(21) link the decision variables α_{ijk} and β_{ijdk}. Constraints (19) state that if a trip of drone d on truck k starts from node $i \in O_-$ and ends at $j \in O_+$, then truck k should also travel from node $i \in O_-$ to node $j \in O_+$. Constraints (20) require that if drone d on truck k leaves truck k at node i to serve customer c, truck k must access node i. Constraints (21) ensure that if drone d on truck k back to truck k at node j, truck k must cross node j. Constraints (22) limit the total goods weight transported on the route of truck group k to not higher than the maximum load capacity of truck k. Constraints (23) indicate that if truck group k is used, then truck k should go through at least one route network node before returning the depot. Constraints (24) state whether or not truck group k is used. Constraints (25) and (26) are subtour elimination constraints. These variables are defined in Constraints (27)–(30).

4.3. Linearization for the model

The last part of the objective function of the above model is nonlinear, so we need to linearize it. The decision variables τ_{ck}^{1+} and τ_{ck}^{1-}, $c \in C, k \in K$ are defined. The last part of the objective function $\sum_{k\in K}\sum_{c\in C}s^T\left(\delta_{ck} - e_c\sum_{d\in D}\sum_{i\in O_-}\beta_{icdk}\right)^+$ can be reformulated as $\sum_{k\in K}\sum_{c\in C}s^T\tau_{ck}^{1+}$ by adding some constraints.

$$\delta_{ck} - e_c\sum_{d\in D}\sum_{i\in O_-}\beta_{icdk} = \tau_{ck}^{1+} - \tau_{ck}^{1-} \quad c \in C, k \in K \tag{31}$$

$$\tau_{ck}^{1+}, \tau_{ck}^{1-} \geq 0 \quad c \in C, k \in K \tag{32}$$

According to the new decision variables and constraints, model M1 is represented as an MILP model.

$$[\text{M2}]\ \text{Minimize}\left\{\sum_{k\in K}s^G\varepsilon_k + \sum_{k\in K}\sum_{i\in O_-}\sum_{j\in O_+}s^K t_{ij}^K\alpha_{ijk} + \sum_{k\in K}\sum_{d\in D}\sum_{i\in O_-}\sum_{c\in C}s^D t_{ic}^D\beta_{icdk} + \sum_{k\in K}\sum_{d\in D}\sum_{j\in O_+}\sum_{c\in C}s^D t_{jc}^D\beta_{cjdk} + \sum_{k\in K}\sum_{c\in C}s^T\tau_{ck}^{1+}\right\} \tag{33}$$

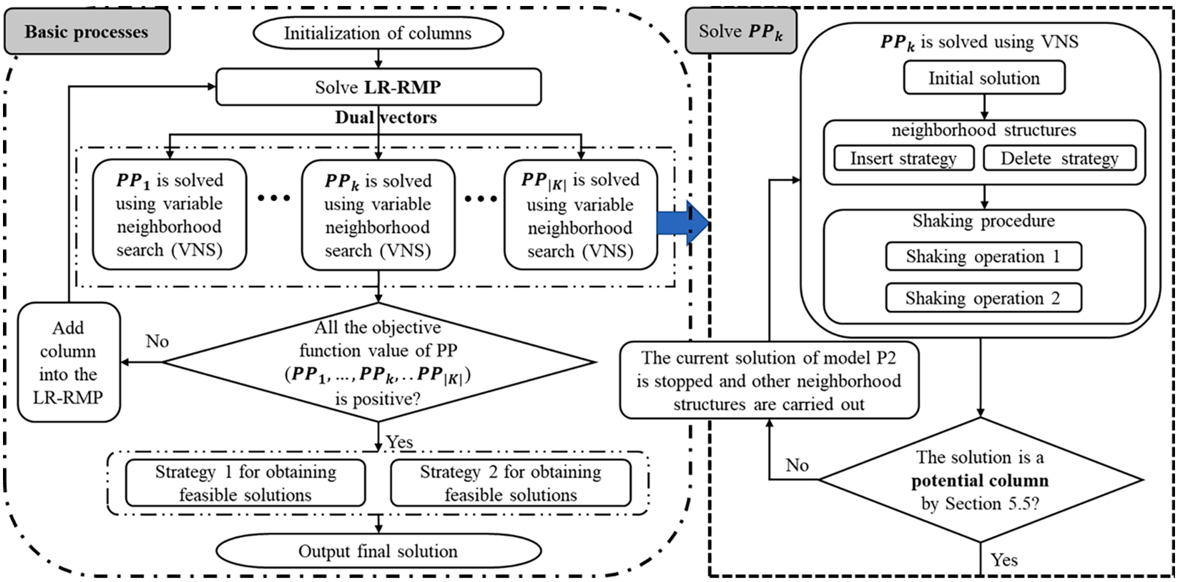


Fig. 1. Flow chart of the column generation-based heuristic algorithm.

subject to: Constraints (2)–(32).

5. Column generation-based heuristic algorithm

For solving the above model under large-scale instances, this paper has developed a column generation-based heuristic algorithm. Some acceleration techniques are also implemented so as to shorten the solution time. Fig. 1 is the flow chart of the column generation-based heuristic algorithm.

5.1. Set partitioning based model

As per the usual practice of the column generation, the original problem described in Section 4 is first framed as a set partitioning model, namely the master problem (MP). R is defined as the set of all possible routes for truck groups. For any feasible route $r \in R$, we define parameters x_{ijr} for truck route, y_{ijdr} for drone route, λ for the usage of the truck group, and ρ_{ir} is the time which is arriving at node i for the truck group in route r. x_{ijr}, y_{ijdr} and λ are binary parameters, while the truck travels from node $i \in O_-$ to node $j \in O_+$ in route r, we have $x_{ijr} = 1$; if drone d on the truck travels from node $i \in O_- \cup C$ to node $j \in O_+ \cup C$ in route r, we have $y_{ijdr} = 1$; and if truck group is used, we have $\lambda = 1$. The cost of route r is denoted by c_r. The cost calculation formula of route r is as follows.

$$c_r = s^G\lambda + s^K\sum_{i\in O_-}\sum_{j\in O_+}t_{ij}^K x_{ijr} + s^D\sum_{d\in D}\sum_{i\in O_-}\sum_{c\in C}t_{ic}^D y_{icdr} + s^D\sum_{d\in D}\sum_{j\in O_+}\sum_{c\in C}t_{jc}^D y_{cjdr} + \sum_{c\in C}s^T\left(\rho_{cr} - e_c\sum_{d\in D}\sum_{i\in O_-}y_{icdr}\right)^+ \tag{34}$$

Let ξ_r be a binary variable, while route r is selected, $\xi_r = 1$. On account of MP, we define R' is the subset of all feasible routes, $R' \subseteq R$, and ξ_r is relaxed as continuous variable. Thus, a linear relaxation of MP (LR-RMP) is formulated. An initial feasible solution can be obtained by designing a heuristic, and the elements of R' are obtained, as described in Section 5.3. Here is LR-RMP:

$$[\mathbf{LR-RMP}]\ \text{Minimize}\sum_{r\in R'}c_r\xi_r \tag{35}$$

subject to

$$\sum_{r\in R'}\sum_{i\in O_-\cup C}\sum_{d\in D}y_{ijdr}\xi_r = 1 \quad j \in C \tag{36}$$

$$\sum_{r\in R'}\xi_r \leq |K| \tag{37}$$

$$\xi_r \geq 0\ r \in R' \tag{38}$$

Objective (35) minimizes the used routes cost. Constraints (36) mean that customers can only be served by drones, and one customer can only be served by one drone. Constraints (37) ensure that at most $|K|$ routes are selected. These variables are defined in Constraints (38).

We add the LR-RMP's dual variables into the pricing problem (PP)for generating a new feasible column. Define dual variables θ_j and π for Constraint (36) and (37), respectively.

5.2. The model of the PP

Because of the PP's feature, it can be decomposed into $|K|$ pricing subproblems, each of which is denoted by PP$_k$. The model PP$_k$ corresponds to truck group k.

$$[\mathbf{PP}\boldsymbol{k}]\text{Minimize}\sigma_k = c_r - \sum_{j\in C}\sum_{i\in O_-\cup C}\sum_{d\in D}\beta_{ijd}\theta_j - \pi \tag{39}$$

subject to

$$\sum_{j\in O_+}\alpha_{0j} = 1 \tag{40}$$

$$\sum_{i\in O_-}\alpha_{i(o+1)} = 1 \tag{41}$$

$$\sum_{i\in O_-}\alpha_{ij} = \sum_{i\in O_+}\alpha_{ji} \leq 1\ j \in O_0 \tag{42}$$

$$\sum_{j\in O_+\cup C}\beta_{0jd} = 1 \quad d \in D \tag{43}$$

$$\sum_{i \in O_- \cup C} \beta_{i(o+1)d} = 1 \;\; d \in D \tag{44}$$

$$\sum_{i \in O_- \cup C} \beta_{ijd} = \sum_{i \in O_+ \cup C} \beta_{jid} \leq 1 j \in O_0 \cup C, d \in D \tag{45}$$

$$\sum_{i \in O_-} \sum_{d \in D} \beta_{icd} \leq 1 \quad c \in C \tag{46}$$

$$\sum_{j \in O_+} \sum_{d \in D} \beta_{cjd} \leq 1 \;\; c \in C \tag{47}$$

$$\beta_{ijd} = 0 \quad i \in C, j \in C, d \in D \tag{48}$$

$$\sum_{c \in C} \beta_{icd} \leq 1 \quad i \in O_-, d \in D \tag{49}$$

$$\sum_{c \in C} \beta_{cjd} \leq 1 \quad j \in O_+, d \in D \tag{50}$$

$$\delta_0 = 0 \tag{51}$$

$$\delta_j \geq \delta_i + t_{ij}^K - M\left(1 - \alpha_{ij}\right) \; i \in O_-, j \in O_+ \tag{52}$$

$$\delta_c \geq \delta_i + t_{ic}^D - M(1 - \beta_{icd}) \; i \in O_-, c \in C, d \in D \tag{53}$$

$$\delta_j \geq \delta_c + t_{jc}^D - M\left(1 - \beta_{cjd}\right) \; c \in C, j \in O_+, d \in D \tag{54}$$

$$\delta_j \geq \delta_i + t_{ij}^K - M\left(1 - \beta_{ijd}\right) \; i \in O_-, j \in O_+, d \in D \tag{55}$$

$$\delta_j \geq \delta_i - M\left(2 - \beta_{icd} - \beta_{cjd}\right) \; c \in C, i \in O_-, j \in O_+, d \in D \tag{56}$$

$$\beta_{ijd} \leq \alpha_{ij} \quad i \in O_-, j \in O_+, d \in D \tag{57}$$

$$\beta_{icd} \leq \sum_{j \in O_+} \alpha_{ij} \quad c \in C, i \in O_-, d \in D \tag{58}$$

$$\beta_{cjd} \leq \sum_{i \in O_-} \alpha_{ij} \;\; j \in O_+, c \in C, d \in D \tag{59}$$

$$\sum_{i \in O_-} \sum_{c \in C} \sum_{d \in D} q_c \beta_{icd} \leq m^K \tag{60}$$

$$\alpha_{0(o+1)} = 1 - \varepsilon \tag{61}$$

$$\varepsilon \geq \beta_{icd} \quad c \in C, i \in O_-, d \in D \tag{62}$$

$$\varphi_{id} - \varphi_{jd} \leq (|O| + |C|)\left(1 - \beta_{ijd}\right) - 1 i \in O_- \cup C, j \in O_+ \cup C, d \in D \tag{63}$$

$$\mu_i - \mu_j \leq |O|\left(1 - \alpha_{ij}\right) - 1 i \in O_-, j \in O_+ \tag{64}$$

$$\delta_i \geq 0 \quad i \in N \tag{65}$$

$$0 \leq \varphi_{id} \leq |O| + |C| \quad i \in N, d \in D \tag{66}$$

$$0 \leq \mu_i \leq |O| \quad i \in N \tag{67}$$

$$\alpha_{ij}, \beta_{ijd}, \varepsilon \in \{0, 1\} \quad i, j \in N, d \in D \tag{68}$$

$$c_r = s^G \varepsilon + s^K \sum_{i \in O_-} \sum_{j \in O_+} t_{ij}^K \alpha_{ij} + s^D \sum_{d \in D} \sum_{i \in O_-} \sum_{c \in C} t_{ic}^D \beta_{icd} + s^D \sum_{d \in D} \sum_{j \in O_+} \sum_{c \in C} t_{jc}^D \beta_{cjd} + \sum_{c \in C} s^T \left(\rho_{cr} - e_c \sum_{d \in D} \sum_{i \in O_-} \beta_{icd}\right)^+ \tag{69}$$

Objective (39) minimizes the reduced cost. Constraints (40)–(45) and (48)–(68) are similar to Constraints (2)–(7) and (10)–(30), respectively. Constraints (46) and (47) guarantee that all customers should be served once. The route cost is calculated in the Constraints (69). Since $\sum_{c\in C} s^T\left(\rho_{cr} - e_c\sum_{d\in D}\sum_{i\in O_-}\beta_{icd}\right)^+$ is nonlinear, it is linearized based on the method introduced in Section 4.3. We define the decision variables τ_c^{2+} and τ_c^{2-}, $c \in C$. $\sum_{c\in C} s^T\left(\rho_{cr} - e_c\sum_{d\in D}\sum_{i\in O_-}\beta_{icd}\right)^+$ can be reformulated as $\sum_{c\in C} s^T\tau_c^{2+}$, where $\tau_c^{2+} \geq 0$, $\tau_c^{2-} \geq 0$ and $\rho_{cr} - e_c\sum_{d\in D}\sum_{i\in O_-}\beta_{icd} = \tau_c^{2+} - \tau_c^{2-}$, $c \in C$.

5.3. Generate initial solution

An initial solution is requiring in the column generation algorithm, which are obtained using a greedy heuristic, the process of which follows the following steps:

Step 1: Given the customer demands increasing order, sort the customers. Initialize a truck group route that is empty.

Step 2: According to the above customer sorted sequence, we assign each customer to a truck group sequentially until assign all customers; in the assignment process, the truck group capacity constraints must be satisfied.

Step 3: The customers assigned to each truck group in Step 2 are evenly assigned to the drones in this truck group according to the customer time windows. Record $m_{dk}, d \in D, k \in K$ as the amount of customers who are assigned to each drone d on truck k, and the customer indexes assigned to each drone.

Step 4: Calculate $\max\limits_{d\in D}\{m_{dk}\}, k \in K$. Initialize the route of each truck group k, containing at least $\max\limits_{d\in D}\{m_{dk}\}$ road network nodes. For each customer, the nearest road network nodes to the customer are selected. These road network nodes are the nodes that trucks must visit. Then we use a greedy algorithm to obtain the truck routes.

Step 5: According to the truck route in Step 4 and the customer assignment in Step 3, each drone's flying route for serving a customer is formed as follows: the nearest road network node to the customer is the node from which the drone returns to the truck, and the previous node on a truck's route is the node where the drone is launched from the truck. When truck group routes are determined, we can obtain the variables values such as the α_{ijk}, β_{ijdk} and ε_k.

Step 6: We solve model P0 by using CPLEX with the computed α_{ijk}, β_{ijdk} and ε_k values. After solving the model P0, the values of the remaining variables are also determined.

Newly added decision variables:

δ_{ik}^K time truck k arrives at node i, $i \in N$;

δ_{ik}^D time a drone on truck k arrives at node i, $i \in N$;

$$\delta_{jk} = \max\left\{\delta_{ik}^K, \delta_{ik}^D\right\}$$

The model for P0 is then as follows:

$$[\mathbf{P}0]\ \text{Minimize}\left\{\sum_{k\in K} s^G\varepsilon_k + \sum_{k\in K}\sum_{i\in O_-}\sum_{j\in O_+} s^K t_{ij}^K\alpha_{ijk} + \sum_{k\in K}\sum_{d\in D}\sum_{i\in O_-}\sum_{c\in C} s^D t_{ic}^D\beta_{icdk} + \sum_{k\in K}\sum_{d\in D}\sum_{j\in O_+}\sum_{c\in C} s^D t_{jc}^D\beta_{cjdk} + \sum_{k\in K}\sum_{c\in C} s^T\left(\delta_{ck} - e_c\sum_{d\in D}\sum_{i\in O_-}\beta_{icdk}\right)^+\right\} \tag{70}$$

subject to

$$\delta_{0k} = 0\ k \in K \tag{71}$$

$$\delta_{0k}^K = 0\ k \in K \tag{72}$$

$$\delta_{0k}^D = 0 \quad k \in K \tag{73}$$

$$\delta_{jk}^K \geq \delta_{ik}^K + t_{ij}^K - M\left(1 - \alpha_{ijk}\right)\ i \in O_-, j \in O_+, k \in K \tag{74}$$

$$\delta_{jk}^K \leq \delta_{ik}^K + t_{ij}^K + M\left(1 - \alpha_{ijk}\right)\ i \in O_-, j \in O_+, k \in K \tag{75}$$

$$\delta_{ck}^D \geq \delta_{ik}^D + t_{ic}^D - M(1 - \beta_{icdk})\ i \in O_-, c \in C, d \in D, k \in K \tag{76}$$

$$\delta_{jk}^D \geq \delta_{ck}^D + t_{jc}^D - M\left(1 - \beta_{cjdk}\right)\ c \in C, j \in O_+, k \in K, d \in D \tag{77}$$

$$\delta_{jk}^D \geq \delta_{ik}^D + t_{ij}^K - M\left(2 - \beta_{ijdk} - \alpha_{ijk}\right) i \in O_-, j \in O_+, k \in K, d \in D \tag{78}$$

$$\delta_{jk}^D \geq \delta_{ik}^D - M\left(2 - \beta_{icdk} - \beta_{cjdk}\right)\ c \in C, i \in O_-, j \in O_+, k \in K, d \in D \tag{79}$$

$$\delta_{ik} = \max\left\{\delta_{ik}^K, \delta_{ik}^D\right\} \quad i \in N, k \in K \tag{80}$$

$$\delta_{ik}, \delta_{ik}^K, \delta_{ik}^D \geq 0\ i \in N, k \in K \tag{81}$$

Objective (70) minimizes the total operational cost, including truck travel cost, drone travel cost, using truck groups fixed cost, and potential penalty for late delivery at customer locations. Constraints (71)–(73) state that the time truck k and drone d depart from the depot is zero. Constraints (74) and (75) compute the travel time of trucks, and Constraints (76)–(79) determine the travel time of drones. Constraints (80) enforce the time truck group k to arrive at node i. Constraints (81) define the domains of variables. The linearization of $\sum_{k\in K}\sum_{c\in C}s^T\left(\delta_{ck}-e_c\sum_{d\in D}\sum_{i\in O_-}\beta_{icdk}\right)^+$ is the same as in Section 4.3.

5.4. Solving the PP

In column generation algorithms, the MP is a set covering problem, with variables associated with vehicle routes and times, which are generated by the PPs. Each PP is NP-hard. We use variable neighborhood search (VNS) to solve PP in this paper. The main idea of VNS is to generate an initial solution firstly, and then try to improve it through neighborhood search. When the solution hasn't been improved for a certain number of consecutive iterations, a perturbation process is performed, which can stop the algorithm from falling into a local optimum. The quality of the initial solution in this paper can be improved by embedding a mathematical model in the VNS algorithm.

The PP involves the decision of truck route and drone route. We first formulate a mathematical model P1 to generate truck route including some fixed road network nodes. Let RR be the set of road network nodes that the truck must visit, $RR \subseteq O$. If $i \in RR$, the truck must through node i.

Decision variables:

α^1_{ij} binary, if the truck travels from node $i \in O_-$ to node $j \in O_+$, $\alpha^1_{ij} = 1$;

μ^1_i integer, if the truck starts from the depot, the μ^1_i th node to be visited is node i;

δ^1_i continuous nonnegative, the time for the truck group arriving at node i.

The model for P1 is as follows:

$$[\mathbf{P}1]\ \text{Minimize}\left\{\sum_{i\in O_-}\sum_{j\in O_+}s^K t^K_{ij}\alpha^1_{ij}\right\} \tag{82}$$

subject to

$$\sum_{j\in O_+}\alpha^1_{0j} = 1 \tag{83}$$

$$\sum_{i\in O_-}\alpha^1_{i(o+1)} = 1 \tag{84}$$

$$\sum_{i\in O_-}\alpha^1_{ij} = \sum_{i\in O_+}\alpha^1_{ji}\ \ j\in O_0 \tag{85}$$

$$\sum_{i\in O_-}\alpha^1_{ij} = \sum_{i\in O_+}\alpha^1_{ji} = 1\ \ j\in RR \tag{86}$$

$$\sum_{i\in O_-}\alpha^1_{ij} = \sum_{i\in O_+}\alpha^1_{ji} = 0\ \ j\notin RR \tag{87}$$

$$\delta^1_0 = 0 \tag{88}$$

$$\delta^1_j \geq \delta^1_i + t^K_{ij} - M\left(1-\alpha^1_{ij}\right)\ \ i\in O_-, j\in O_+ \tag{89}$$

$$\delta^1_j \leq \delta^1_i + t^K_{ij} + M\left(1-\alpha^1_{ij}\right)\ \ i\in O_-, j\in O_+ \tag{90}$$

$$\mu^1_i - \mu^1_j \leq |O|\left(1-\alpha^1_{ij}\right) - 1\, i\in O_-, j\in O_+ \tag{91}$$

$$\delta^1_i \geq 0 \quad i\in N \tag{92}$$

$$0 \leq \mu^1_i \leq |O| \quad i\in N \tag{93}$$

$$\alpha^1_{ij} \in \{0,1\} \quad i,j\in N \tag{94}$$

Constraints (83) and (84) guarantee that truck k initially leaves the depot and ends up back at the depot. Constraints (85) are flow conservation constraints. Constraints (86) and (87) guarantee that if $j \in RR$, then the truck must visit road network node j; otherwise, the truck does not visit road network node j. Constraints (88) indicate the truck group time leaves the depot is zero. Constraints (89) and (90) are truck travel time constraints. Constraints (91) are subtour elimination constraints. These variables are defined in

Constraints (92)–(94).

Then, based on the truck route obtained by P1, another mathematical model P2 is formulated to yield drone routes. The 0–1 parameter a_{ij} is defined to represent the truck route, if the truck travels from node $i \in O_-$ to node $j \in O_+$, $a_{ij} = 1$. Define the RD as customer nodes set that drones on the truck must visit, $RD \subseteq C$. If $i \in RD$, the drone must visit node i.

Decision variables:

β^2_{ijd} binary, if drone d travels from node $i \in O_- \cup C$ to node $j \in O_+ \cup C$, $\beta^2_{ijd} = 1$;

φ^2_{id} integer, if drone d starts from the depot, the φ^2_{id} th node to be visited is node i;

δ^2_i continuous nonnegative, time for the truck group at node i.

The model for P2 is as follows:

$$[\mathbf{P2}]\ \text{Minimize}\left\{\sum_{i\in O_-}\sum_{j\in O_+} s^K t^K_{ij} a_{ij} + \sum_{d\in D}\sum_{i\in O_-}\sum_{c\in C} s^D t^D_{ic}\beta^2_{icd} + \sum_{d\in D}\sum_{j\in O_+}\sum_{c\in C} s^D t^D_{jc}\beta^2_{cjd} + \sum_{c\in C} s^T\left(\delta^2_c - e_c\sum_{d\in D}\sum_{i\in O_-}\beta^2_{icd}\right)^+ + s^G - \sum_{j\in C}\sum_{i\in O_-\cup C}\sum_{d\in D}\beta^2_{ijd}\theta_j - \pi\right\} \tag{95}$$

subject to

$$\sum_{j\in O_+\cup C}\beta^2_{0jd} = 1 \quad d\in D \tag{96}$$

$$\sum_{i\in O_-\cup C}\beta^2_{i(o+1)d} = 1 \quad d\in D \tag{97}$$

$$\sum_{i\in O_-\cup C}\beta^2_{ijd} = \sum_{i\in O_+\cup C}\beta^2_{jid} j\in O_0\cup C, d\in D \tag{98}$$

$$\sum_{i\in O_-}\sum_{d\in D}\beta^2_{icd} = 1 \quad c\in RD \tag{99}$$

$$\sum_{j\in O_+}\sum_{d\in D}\beta^2_{cjd} = 1 \quad c\in RD \tag{100}$$

$$\sum_{i\in O_-}\sum_{d\in D}\beta^2_{icd} = 0 \quad c\notin RD \tag{101}$$

$$\sum_{j\in O_+}\sum_{d\in D}\beta^2_{cjd} = 0 \quad c\notin RD \tag{102}$$

$$\beta^2_{ijd} = 0 \quad i\in C, j\in C, d\in D \tag{103}$$

$$\sum_{c\in C}\beta^2_{icd} \le 1 \quad i\in O_-, d\in D \tag{104}$$

$$\sum_{c\in C}\beta^2_{cjd} \le 1 \quad j\in O_+, d\in D \tag{105}$$

$$\delta^2_0 = 0 \tag{106}$$

$$\delta^2_j \ge \delta^2_i + t^K_{ij} - M(1-a_{ij}) \ i\in O_-, j\in O_+ \tag{107}$$

$$\delta^2_c \ge \delta^2_i + t^D_{ic} - M(1-\beta^2_{icd}) \ i\in O_-, c\in C, d\in D \tag{108}$$

$$\delta^2_j \ge \delta^2_c + t^D_{jc} - M\left(1-\beta^2_{cjd}\right) \ c\in C, j\in O_+, d\in D \tag{109}$$

$$\delta^2_j \ge \delta^2_i + t^K_{ij} - M\left(1-\beta^2_{ijd}\right) \ i\in O_-, j\in O_+, d\in D \tag{110}$$

$$\delta^2_j \ge \delta^2_i + t^K_{ij} - M\left(2-a_{ij}-\beta^2_{ijd}\right) i\in O_-, j\in O_+, d\in D \tag{111}$$

$$\delta^2_j \ge \delta^2_i - M\left(2-\beta^2_{icd}-\beta^2_{cjd}\right) c\in C, i\in O_-, j\in O_+, d\in D \tag{112}$$

$$\beta^2_{ijd} \le a_{ij} \quad i\in O_-, j\in O_+, d\in D \tag{113}$$

$$\beta_{icd}^2 \le \sum_{j \in O_+} a_{ij} \quad i \in O_-, c \in C, d \in D \tag{114}$$

$$\beta_{cjd}^2 \le \sum_{i \in O_-} a_{ij} \quad c \in C, j \in O_+, d \in D \tag{115}$$

$$\sum_{i \in O_-} \sum_{c \in C} \sum_{d \in D} q_c \beta_{icd}^2 \le m^K \tag{116}$$

$$\varphi_{id}^2 - \varphi_{jd}^2 \le (|O| + |C|)\left(1 - \beta_{ijd}^2\right) - 1 i \in O_- \cup C, j \in O_+ \cup C, d \in D \tag{117}$$

$$0 \le \varphi_{id}^2 \le |O| + |C| \quad i \in N, d \in D \tag{118}$$

$$\delta_i^2 \ge 0 \quad i \in N \tag{119}$$

$$\beta_{ijd}^2 \in \{0, 1\} \, i, j \in N, d \in D \tag{120}$$

Objective (95) minimizes the reduced cost. Constraints (96) and (97) ensure that each drone initially leaves the depot and ends up back at the depot after completing all distribution tasks. Constraints (98) are flow conservation constraints. Constraints (99)–(102) guarantee that if $j \in RD$, then the drone services customer j; otherwise the drone does not serve customer j. Constraints (103) require that the drone serves at most one customer after each departure from the truck. Constraints (104) and (105) indicate that drone d leaves the truck at most once at node i and returns to the truck at most once at node j. Constraints (106) ensure the truck group initially leaves the depot. Constraints (107) control the trucks travel time. Constraints (108)–(112) calculate the travel time. Constraints (108) ensure that the time for the truck group to leave customer c should not be less than the time for the truck group to leave node i, plus drone travel time from node i to customer c. Constraints (109) require the truck group time to leave node j is not less than the time for the truck group to leave customer c, plus drone travel time from customer c to node j. Constraints (110) and (111) indicate if the truck group starts from node i and arrives at node j, then the time at which the truck group leaves node j should be no less than the time at which the truck group leaves node i, plus truck travel time from node i to node j. Constraints (112) guarantee if drone arrives at node j after starting from node i to serve customer c, then the time for the truck group to leave node j should not be earlier than the time for the truck group to leave node i. Constraints (113) state that if the drone starts from node $i \in O_-$ to $j \in O_+$, the truck should also start from node $i \in O_-$ to $j \in O_+$. Constraints (114) mean that if drone d on the truck leaves the truck from node i to serve customer c, the truck must access node i. Constraints (115) ensure that if drone d on the truck returns to the truck at node j, the truck must cross node j. Constraints (116) limit the total goods weight transported on the route of a truck group to not higher than the truck maximum load capacity. Constraints (117) are subtour elimination constraints. These variables are defined in Constraints (118)–(120).

5.4.1. *Initial solution*

As already mentioned, the VNS algorithm for solving the PP requires an initial solution, which is generated as following.

Step 1: According to the increasing order of θ_j, $j \in C$, sort customers.

Step 2: Based on the above sorted sequence of customers and the truck group capacity constraints, customers are assigned to a truck group until its capacity is reached.

Step 3: For each customer that is assigned in **Step 2**, the nearest road network node to the customer is selected.

Step 4: The model P1 is solved by CPLEX for obtaining the truck route.

Step 5: Given the truck route obtained in **Step 4,** the model P2 is solved by CPLEX for obtaining the routes of the drones.

5.4.2. *Neighborhood structure*

The decisions of the PP include the truck and drone routes, which are relatively complex compared with the conventional routing optimization. In this paper, the insert and delete neighborhood structure is designed by considering the specific characteristics of the established model.

Insert strategy: Insertion of the road network node in the truck route

This strategy is related to the operation on α_{ij}. Select road network node i and determine whether this node is visited in the truck route. If not, add it to the truck route. Then the mathematical model P1 is solved by CPLEX to obtain the truck route. Next, the obtained objective value F_1 is compared with the objective value F_0 obtained by solving LR-RMP last time. If $F_1 < F_0$, the mathematical model P2 is solved by CPLEX to obtain the truck group routes. If the objective value of P2 $F_2 < 0$, there will generate a new column which is added to LR-RMP; otherwise, no new column will be generated, and the process continues until a stopping criterion is satisfied. Repeat this operation until all road network nodes have been handled by the insert strategy.

Delete strategy: Deletion of a road network node in the truck route

The deletion of a road network node in the truck route is mainly to change α_{ij}. Select road network node i and determine whether this node is visited in the truck route. If so, remove node i from the truck route. Then the model P1 is solved by CPLEX to obtain the truck route. Next, the obtained objective value F_1 is compared with the objective value F_0 obtained by solving LR-RMP last time. If $F_1 < F_0$, the model P2 is solved by CPLEX to obtain the truck group routes. If the objective value of P2 $F_2 < 0$, there will generate a new column which is added to LR-RMP; otherwise, no new column will be generated. Repeat this operation until all road network node have been handled by the delete strategy.

Insert strategy: Insertion of a customer in a drone route

The customer insertion in the drone route is mainly to change β_{ijd}. The mathematical model P1 is first solved by CPLEX to obtain the truck route. Then, select customer i and determine whether it is visited in a drone route. If not and the dual variable value of this customer is greater than zero, add i to a drone route. The mathematical model P2 is then solved by CPLEX to obtain the truck group routes. If the objective value of P2 $F_2 < 0$, there will generate a new column which is added to LR-RMP; otherwise, no new column will be generated. Repeat this operation until all customers have been handled by the insert strategy.

Delete strategy: Deletion of a customer in a drone route

The customer deletion in a drone route is mainly to change β_{ijd}. The mathematical model P1 is first solved by CPLEX to obtain the truck route. Then, select customer i and determine whether it is visited in a drone route. If so, delete i. The mathematical model P2 is then solved by CPLEX to obtain the truck group routes. If the objective value of P2 $F_2 < 0$, there will generate a new column which is added to LR-RMP; otherwise, no new column will be generated. Repeat this operation until all customers have been handled by the insert strategy.

5.4.3. Shaking procedure

During the neighborhood search, the solution solved by the algorithm may fall into a local optimum. When the current solution of successive iterations is no better than the current optimal solution, we execute a shaking operation, which is addressed as follows.

According to the customers number that the truck group serve, we randomly generate a set of nodes that must be travelled by the truck. Based on the set of nodes, i.e., the input of the previously formulated model P1, we use CPLEX for solving the model P1 and obtain the truck route. The solved objective value is denoted by F_1; and let F_0 be the objective value of the LR-RMP's solution obtained in the previous the column generation procedure iteration. If $F_1 < F_0$, we further use the CPLEX for solving the previously formulated model P2; and the solved objective value is denoted by F_2. If $F_2 < 0$, the newly generated plan is added into the LR-RMP; here the plan is the truck group's plan, which include the truck route and its carried drones' routes. If $F_2 \geq 0$, the newly generated plan is not added into the LR-RMP.

After the above shaking operation, the obtained solution may be worse than the current solution. In this case, the obtained solution can also be used as an initial solution for the next iteration of the neighborhood search.

5.5. Strategies for accelerating of VNS

The PP is solved for several times by column generation-based heuristic algorithm. Saving PP often takes up most of the computation time. It is therefore very important to design some strategies to accelerate the solution of the PP. The data-driven solution method aims to generate a new solution by analyzing, mining, and using historical operation data. The acceleration strategy designed in this section applies the data-driven idea to the algorithm used for solving the PP, which can further shorten the computation time of the PP.

We can obtain new columns by solving the PP. The PP's objective is minimizing the reduced cost. However, the column with the smallest reduced cost may not be effective for improving the final solution quality, so the potential column to improve the final solution quality should be chosen instead of the current optimal column. In the process of solving the PP, we first determine whether it is a potential column on the basis of some criterion for a newly generated column, and if so, we add the column to the LR-RMP. This strategy is a trade-off between time and the solution quality.

Based on the idea of a data-driven algorithm, we fixed the number of truck groups, the number of customers and the location of nodes, and changed other parameters to carry out a number of experiments to get the total cost of truck group cc_k. Define $UB = b\max_{k\in K}\{cc_k\}$, where b is a predefined coefficient and $0 \leq b \leq 1$. When applying VNS to solve PP, the mathematical model P1 is solved by CPLEX to obtain the truck route, and the objective function values F_1 compared with UB. If $F_1 > UB$, stop solving mathematical model P2 to the next iteration. This strategy can improve the efficiency of PP.

5.6. Strategies for obtaining feasible solutions

Strategy 1: Solve the LR-RMP with 0–1 decision variables by CPLEX according to the column set obtained by column generation program. The integer solution is the final solution.

Strategy 2: Since the LR-RMP decision variables value may be non-integer (i.e., ξ_r), we define and calculate the probability ϵ_{ik}, ϵ_{ik} represents the probability that customer i belongs to truck group k. For each truck group k and each customer i, ϵ_{ik}'s value is calculated as $\epsilon_{ik} = \sum_{r\in R}\sum_{j\in O_-\cup C}\sum_{d\in D} y_{jidr}\xi_r$. Define RD_k as the set of customers that truck group k should serve. Based on the decreasing order of the values of $\{\epsilon_{ik}\}_{i\in C,k\in K}$, customers are assigned to a truck group one pair at a time until all customers are assigned to one truck group. The capacity limit of each truck group should not be exceeded in determining the elements of RD_k. For truck group k, the mathematical model P3_k is solved by CPLEX.

$$[\boldsymbol{P}3_k]\ \text{Minimize}\left\{\sum_{i\in O_-}\sum_{j\in O_+} s^K t^K_{ij} a_{ijk} + \sum_{d\in D}\sum_{i\in O_-}\sum_{c\in C} s^D t^D_{ic}\beta_{icd} + \sum_{d\in D}\sum_{j\in O_+}\sum_{c\in C} s^D t^D_{jc}\beta_{cjd} + \sum_{c\in C} s^T\left(\delta_c - e_c\sum_{d\in D}\sum_{i\in O_-}\beta_{icd}\right)^+ + s^G\varepsilon\right\} \quad (121)$$

subject to

Constraints (40)–(44) and (48)–(68)

$$\sum_{i\in O_-\cup C}\beta_{ijd} = \sum_{i\in O_+\cup C}\beta_{jid}\ j\in O_0\cup C, d\in D \tag{122}$$

$$\sum_{i\in O_-}\sum_{d\in D}\beta_{icd} = 1 \quad c\in RD_k \tag{123}$$

$$\sum_{j\in O_+}\sum_{d\in D}\beta_{cjd} = 1 \quad c\in RD_k \tag{124}$$

$$\sum_{i\in O_-}\sum_{d\in D}\beta_{icd} = 0 \quad c\notin RD_k \tag{125}$$

$$\sum_{j\in O_+}\sum_{d\in D}\beta_{cjd} = 0\, c\notin RD_k \tag{126}$$

Objective (121) minimizes the total operational cost, including truck travel cost, drone travel cost, using truck groups fixed cost, and potential penalty for late delivery at customer locations. Constraints (122) are flow conservation constraints. Constraints (123)–(126) guarantee that if $c \in RD_k$, then the drone services customer c; otherwise, the drone does not service customer c.

6. Numerical experiments

To evaluate the algorithmic performance, numerical experiments are carried out. Some management insights are also derived on the basis of a series of sensitivity analyses. All experiments are conducted on a workstation with Intel(R) Xeon(R) CPU E5-2680 v4 @ 2.40 GHz and 256 GB RAM. The algorithms and models (LR-RMP, P0, P1, P2, P3 and PP) are implemented in C# (Visual Studio 2019). The time limit for all instances is 3600 s.

6.1. Experimental settings

As mentioned in Section 1, the CAAC has issued the licenses for the use of drones in delivery to some qualified companies such as the Antwork, which is dedicated to building an urban aerial drone operation network and has implemented some drone delivery practices in Hangzhou. In this study, a 30 km urban circle centered at Hangzhou is selected as the data source of the problem for our experiments. Fig. 2 provides an example of truck groups delivery routes. In Fig. 2, the truck routes are shown as solid lines and the drone routes are shown as dotted lines. There are six truck group routes in Fig. 2, and all red blue and purple dotted lines are drone routes.

According to the technical parameters of the drones used by the Antwork, we set 30 km/h as the average truck speed and set 48 km/h as the drone speed. Based on the study of Wang & Sheu (2019), m^k is 100 wt units, the drone's maximum load capacity is 20 wt units, and the parameter s^G is 100 CNY. Since the capacity of the drone is m^D, the customer needs the drone for service, so the customer's demand does not exceed m^D. We assume that customer demands are between (0, 20] weight units. The study by Wang & Sheu (2019)

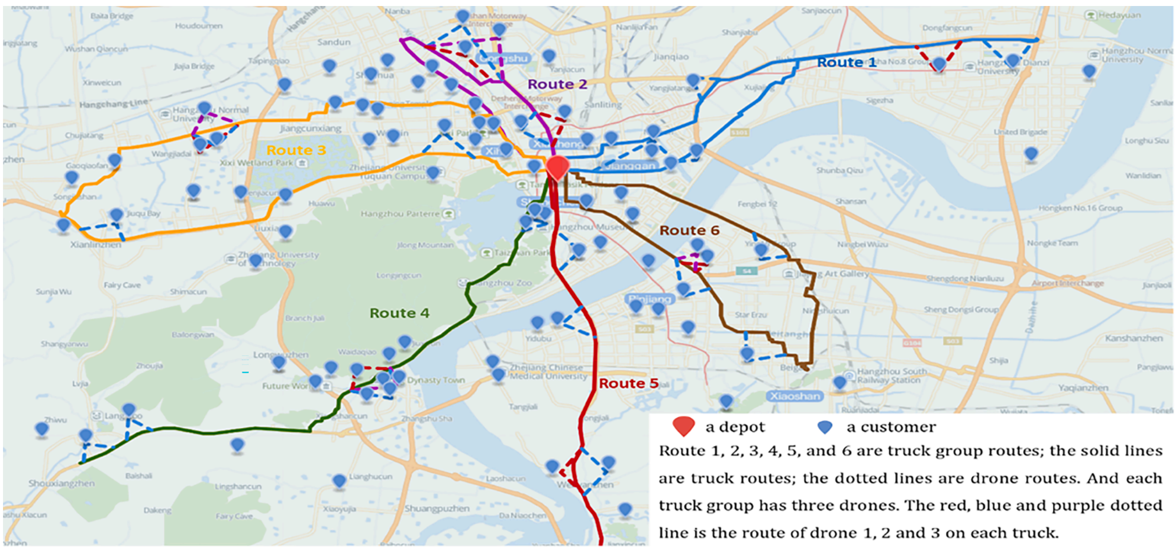


Fig. 2. Six truck groups' routes serving 30 customers.

mentions that the cost of drone is usually 0.19 CNY/km, so we define s^D as 9.12 CNY/h. The truck traveling cost s^K is calculated based on distance travelled, gasoline price and truck speed. We assume that the gasoline price is 8.24 Chinese Yuan (CNY)/L, and the average fuel consumption of a truck is 0.10 L/km. Based on the truck speed, we have s^K is 24.72 CNY/h. We assume that the parameter s^W is 1000 CNY/h. Four groups of problem instances with different sizes are generated. As shown in Table 2, we assume that truck groups K ranges from one to six and customers C ranges from five to 30. Considering the current legal restrictions on drone deliveries in residential areas, customer nodes include locations such as delivery cabinets and customers.

6.2. Experimental results

6.2.1. Comparing strategies for solving problem PP

Here, we assess the effectiveness of these strategies for solving PP. The first strategy is to solve PP by CPLEX, the second strategy is to solve PP using the VNS algorithm, and the third strategy is to solve PP using the accelerated VNS algorithm. Comparative results of the three strategies in Table 3 and 4.

As can be seen from Table 3, the three strategies can efficiently solve the model obtain best solutions in ISG1-ISG3. Because ISG1 instance size is small, the solution time of the first strategy is very short, and the first strategy is significantly faster than the second strategy and the third strategy. In ISG2 andISG3 instances, the first strategy cannot obtain the optimal solution with 3600 s, while the second strategy and the third strategy can obtain the result in a shorter time. In addition, compared with the second strategy, the third strategy can make the PP solution efficiency increases by about 2.93%. Therefore, the third strategy is superior to the second strategy.

As shown in Table 4, the gap between the third strategy and the second strategy is zero. Since the instance size increases, the first strategy solution time is very long, and the result cannot be solved within 3600 s. The second strategy and the third strategy are significantly faster than the first strategy. Compared with the second strategy, the third strategy can make the PP solution efficiency increases by about 5.69%. According to the results of Tables 3 and 4, the third strategy in solution quality and computation time is the best one in the three strategies. In the following experiments, we therefore adopt the third strategy.

6.2.2. Comparing strategies for obtaining feasible solutions

Two strategies are proposed to generate feasible solutions in column generation-based heuristic algorithm. Hence, the objective values and computation time of these strategies are compared by experiments. The strategies are elaborated in Section 5.6. The results under all instances in Tables 5 and 6.

Column Δ_{F_5} in Table 5 indicates that the solution's objective value of strategy 2 is on average 0.06% below that of strategy 1. Column Δ_{t_5} in Table 5 indicates that strategy 2 computation time is on average 3.99% below that of strategy 1. Thus, strategy 2 is superior to strategy 1 on the small-scale instances. In addition, for the results in large-scale instances, column Δ_{F_5} in Table 6 indicates that the solution values of strategy 2 are on average 0.12% below those of strategy 1. However, column Δ_{t_5} in Table 6 indicates that the computation time of strategy 2 is on average 6.16% above that of strategy 1. Since strategy 2 can obtain better solutions than strategy 1 and the computation times of the two strategies are similar, in the following experiments we adopt strategy 2 obtain feasible solutions.

6.2.3. Evaluating the algorithm performance

From the previous series of experiments, we determine the best strategy in each step embedded in the column generation-based algorithm. Here we conduct some experiments for evaluating the proposed algorithm quality. We compare the results of the algorithm with the optimal results of CPLEX in Table 7 and 8.

The Table 7 results show that the heuristic algorithm can obtain a solution whose objective value equals the optimal solution value found by CPLEX in most instances. For the instances that CPLEX cannot solve in one hour, within a short time, the algorithm can obtain good solutions. In Table 8, the results indicate that the algorithm can solve these large-scale instances in one hour and the solutions are better than the incumbent solution obtained by CPLEX in one hour. The results show that the algorithm has good performance in terms of solution quality and computation time.

Another finding is that the proposed algorithm can work in a rolling horizon manner, which means the customer orders are handled in batches. Since the largest instance group (i.e., ISG 6) contains 30 orders and the computation time for the ISG 6 is below about 15 min, our proposed methodology can solve a batch of orders containing about 30 orders every 15 min. In other words, in a realistic context, in which orders emerge and enter the system at an average rate of 120 orders per hour, our proposed methodology can handle the accumulated orders every 15 min.

Table 2
Number of customers and truck groups in each instance group.

Group ID	Customers ($\|C\|$)	Truck groups ($\|K\|$)
ISG1	5	1
ISG2	10	2
ISG3	15	3
ISG4	20	4
ISG5	25	5
ISG6	30	6

Table 3
Comparative effectiveness of three strategies for solving PP (small-scale instances).

Instances		Solution value and time for different strategies for solving PP						Gap					
Scale	ID	F_1	t_1(s)	F_2	t_2(s)	F_3	t_3(s)	Δ_{F_1}	Δ_{t_1}	Δ_{F_2}	Δ_{t_2}	Δ_{F_3}	Δ_{t_3}
ISG1	1	108	11	108	92	108	95	0.00%	736.36%	0.00%	763.64%	0.00%	3.26%
	2	108	8	108	90	108	97	0.00%	1025.00%	0.00%	1112.50%	0.00%	7.78%
	3	108	12	108	92	108	96	0.00%	666.67%	0.00%	700.00%	0.00%	4.35%
	4	108	10	108	92	108	99	0.00%	820.00%	0.00%	890.00%	0.00%	7.61%
	5	108	12	108	93	108	101	0.00%	675.00%	0.00%	741.67%	0.00%	8.60%
ISG2	6	210	526	210	315	210	285	0.00%	−40.11%	0.00%	−45.82%	0.00%	−9.52%
	7	210	579	210	410	210	372	0.00%	−29.19%	0.00%	−35.75%	0.00%	−9.27%
	8	–	>3600	210	360	210	327	–	–	–	–	0.00%	−9.17%
	9	–	>3600	210	251	210	239	–	–	–	–	0.00%	−4.78%
	10	210	264	210	274	210	263	0.00%	3.79%	0.00%	−0.38%	0.00%	−4.01%
ISG3	11	–	>3600	316	296	316	295	–	–	–	–	0.00%	−0.34%
	12	–	>3600	216	496	216	450	–	–	–	–	0.00%	−9.27%
	13	–	>3600	217	529	217	499	–	–	–	–	0.00%	5.67%
	14	–	>3600	316	321	316	263	–	–	–	–	0.00%	−18.07%
	15	–	>3600	217	604	217	571	–	–	–	–	0.00%	−5.46%
Average								0.00%	482.19%	0.00%	515.73%	0.00%	−2.93%

Notes:(1) F_1, F_2, F_3 is the solution of the method for solving PP by CPLEX, the VNS algorithm, the VNS using accelerating strategies, respectively. And t_1, t_2, t_3 is the computation time of the method for solving PP by CPLEX, the VNS algorithm, the VNS with using accelerating strategies, respectively. (2) $\Delta_{F_1} = (F_2 - F_1)/F_1$, $\Delta_{F_2} = (F_3 - F_1)/F_1$, $\Delta_{F_3} = (F_3 - F_2)/F_2$, $\Delta_{t_1} = (t_2 - t_1)/t_1$, $\Delta_{t_2} = (t_3 - t_1)/t_1$, $\Delta_{t_3} = (t_3 - t_2)/t_2$.

Table 4
Comparative effectiveness of three strategies for solving PP (large-scale instances).

Instances		Solution value and time for different strategies for solving PP						Gap					
Scale	ID	F_1	t_1(s)	F_2	t_2(s)	F_3	t_3(s)	Δ_{F_1}	Δ_{t_1}	Δ_{F_2}	Δ_{t_2}	Δ_{F_3}	Δ_{t_3}
ISG4	1	–	>3600	321	359	321	337	–	–	–	–	0.00%	−6.13%
	2	–	>3600	321	830	321	786	–	–	–	–	0.00%	−5.30%
	3	–	>3600	321	545	321	513	–	–	–	–	0.00%	−5.87%
	4	–	>3600	321	420	321	407	–	–	–	–	0.00%	−3.10%
	5	–	>3600	320	390	320	348	–	–	–	–	0.00%	−10.77%
ISG5	6	–	>3600	427	483	427	477	–	–	–	–	0.00%	−1.24%
	7	–	>3600	427	524	427	484	–	–	–	–	0.00%	−7.63%
	8	–	>3600	427	658	427	561	–	–	–	–	0.00%	−14.74%
	9	–	>3600	427	468	427	443	–	–	–	–	0.00%	−5.34%
	10	–	>3600	427	435	427	437	–	–	–	–	0.00%	0.46%
ISG6	11	–	>3600	548	627	548	567	–	–	–	–	0.00%	−9.57%
	12	–	>3600	450	688	450	674	–	–	–	–	0.00%	−2.03%
	13	–	>3600	548	879	548	803	–	–	–	–	0.00%	−8.65%
	14	–	>3600	548	534	548	537	–	–	–	–	0.00%	0.56%
	15	–	>3600	450	798	450	750	–	–	–	–	0.00%	−6.02%
Average								–	–	–	–	0.00%	−5.69%

Notes:(1) F_1, F_2, F_3 is the solution of the method for solving PP by CPLEX, the VNS algorithm, the VNS using accelerating strategies, respectively. And t_1, t_2, t_3 is the computation time of the method for solving PP by CPLEX, the VNS algorithm, the VNS with using accelerating strategies, respectively. (2) $\Delta_{F_1} = (F_2 - F_1)/F_1$, $\Delta_{F_2} = (F_3 - F_1)/F_1$, $\Delta_{F_3} = (F_3 - F_2)/F_2$, $\Delta_{t_1} = (t_2 - t_1)/t_1$, $\Delta_{t_2} = (t_3 - t_1)/t_1$, $\Delta_{t_3} = (t_3 - t_2)/t_2$.

6.3. Sensitivity analyses

Sensitivity analyses are carried out to derive some managerial implications. More specifically, these experiments investigate the influence of some important parameters, which include the number of drones, the number of road network nodes, the speed of drones and the location distribution of customers.

6.3.1. The impact of the number of drones

Drones number may have an impact on the solution cost. In our problem, each truck group carries $|D|$ drones. The sensitivity analysis is conducted on the number of drones per truck group for three instance groups (ISG 3, 4, 5). The results are shown in Fig. 3, which demonstrates that deploying more drones on each truck can reduce cost, but the trend of cost reduction slows down as the drones number increases. Increasing the number of drones on each truck can improve transportation efficiency and reduce reliance on trucks, which is a favorable impact for decision makers and stakeholders as they can complete deliveries faster and provide greater efficiency. But the more drones deployed on each truck, the more drones will need to be purchased and maintained later. In addition,

Table 5
Comparing strategies for obtaining feasible solutions (small-scale instances).

Instances		Solution value and time for strategy 1		Solution value and time for strategy 2		Gap	
Scale	ID	F_4	t_4(s)	F_5	t_5(s)	Δ_{F_5}	Δ_{t_5}
ISG1	1	108	102	108	95	0.00%	−6.86%
	2	108	95	108	97	0.00%	2.11%
	3	108	95	108	96	0.00%	1.05%
	4	108	93	108	99	0.00%	6.45%
	5	108	95	108	101	0.00%	6.32%
ISG2	6	210	325	210	285	0.00%	−12.31%
	7	210	429	210	372	0.00%	−13.29%
	8	210	397	210	327	0.00%	−17.63%
	9	210	270	210	239	0.00%	−11.48%
	10	210	306	210	263	0.00%	−14.05%
ISG3	11	316	303	316	295	0.00%	−2.64%
	12	218	473	216	450	−0.92%	−4.86%
	13	217	482	217	499	0.00%	3.53%
	14	316	255	316	263	0.00%	3.14%
	15	217	567	217	571	0.00%	0.71%
Average						−0.06%	−3.99%

Notes: (1) F_4 and F_5 represent the MILP model objective value by column generation-based heuristic algorithm with strategy 1 for obtaining feasible solutions and with strategy 2 for obtaining feasible solutions, respectively. And t_4 and t_5 denote the computation time of algorithm with strategy 1 for obtaining feasible solutions and with strategy 2 for obtaining feasible solutions to solve the MILP model, respectively. (2) $\Delta_{F_5} = (F_5 - F_4)/F_4$. (5) $\Delta_{t_5} = (t_5 - t_4)/t_4$.

Table 6
Comparing strategies for obtaining feasible solutions (large-scale instances).

Instances		Solution value and time for strategy 1		Solution value and time for strategy 2		Gap	
Scale	ID	F_4	t_4(s)	F_5	t_5(s)	Δ_{F_5}	Δ_{t_5}
ISG4	1	321	286	321	337	0.00%	17.83%
	2	321	760	321	786	0.00%	3.42%
	3	322	505	321	513	−0.31%	1.58%
	4	321	355	321	407	0.00%	14.65%
	5	321	311	320	348	−0.31%	11.90%
ISG5	6	427	432	427	477	0.00%	10.42%
	7	428	467	427	484	−0.23%	3.64%
	8	427	569	427	561	0.00%	−1.41%
	9	427	410	427	443	0.00%	8.05%
	10	427	422	427	437	0.00%	3.55%
ISG6	11	548	562	548	567	0.00%	0.89%
	12	452	624	450	674	−0.44%	8.01%
	13	548	783	548	803	0.00%	2.55%
	14	548	535	548	537	0.00%	0.37%
	15	452	701	450	750	−0.44%	6.99%
Average						−0.12%	6.16%

Notes: (1) F_4 and F_5 represent the MILP model objective value by column generation-based heuristic algorithm with strategy 1 for obtaining feasible solutions and with strategy 2 for obtaining feasible solutions, respectively. And t_4 and t_5 denote the computation time of algorithm with strategy 1 for obtaining feasible solutions and with strategy 2 for obtaining feasible solutions to solve the MILP model, respectively. (2) $\Delta_{F_5} = (F_5 - F_4)/F_4$. (5) $\Delta_{t_5} = (t_5 - t_4)/t_4$.

the deployment of more drones may require technical adaptation of trucks in order to mount and operate drones on trucks. Therefore, with more drones, decision makers have more choices and are more likely to plan a better delivery scheme, but decision makers should consider numerous factors to determine the number of drones on each truck.

6.3.2. The impact of the number of nodes in the road network

Road network nodes are nodes of constructing the truck routes, through which the trucks travel; a separation node is a candidate location where a truck launch a drone, while a confluence node is a candidate location where a drone is returned back to a truck. The number of road network nodes may have an impact on the solution cost. Hence, some experiments are conducted on three instance groups (ISG 3, 4, 5). In Fig. 4, the results show that when more nodes are considered in the road network, the total cost can be reduced, but not significantly. Adding road network nodes can improve regional accessibility and connectivity, and by increasing the number of nodes, decision makers can better plan the flow of goods and choose shorter paths and more suitable node connections. However,

Table 7
Comparing between CPLEX and column generation-based algorithm (small-scale instances).

Instances		CPLEX		Column generation		
Scale	ID	F_{CPLEX}	$t_{CPLEX}(s)$	F_{CG}	$t_{CG}(s)$	$\Delta_{F_{CG}}$
ISG1	1	108	1	108	95	0.00%
	2	108	1	108	97	0.00%
	3	108	2	108	96	0.00%
	4	108	1	108	99	0.00%
	5	108	2	108	101	0.00%
ISG2	6	210	32	210	285	0.00%
	7	210	74	210	372	0.00%
	8	210	37	210	327	0.00%
	9	210	34	210	239	0.00%
	10	210	38	210	263	0.00%
ISG3	11	316	>3600	316	295	0.00%
	12	216	>3600	216	450	0.00%
	13	216	>3600	217	499	0.46%
	14	316	>3600	316	263	0.00%
	15	216	>3600	217	571	0.46%
Average						0.06%

Notes: (1) F_{CPLEX} and F_{CG} represent the MILP model objective value by CPLEX and column generation-based heuristic algorithm, respectively. And t_{CPLEX} and t_{CG} denote the computation time of CPLEX and algorithm to solve the MILP model, respectively. F_{CPLEX} corresponding to $t_{CPLEX} > 3600$ in the table is the MILP model objective value by CPLEX for $t_{CPLEX} = 3600$. (2) $\Delta_{F_{CG}} = (F_{CG} - F_{CPLEX})/F_{CPLEX}$.

Table 8
Comparing between CPLEX and column generation-based algorithm (large-scale instances).

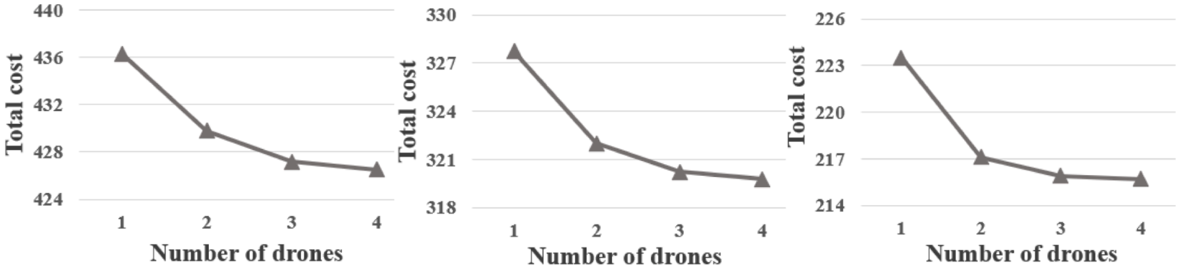
Instances		CPLEX		Column generation		
Scale	ID	F_{CPLEX}	$t_{CPLEX}(s)$	F_{CG}	$t_{CG}(s)$	$\Delta_{F_{CG}}$
ISG4	1	325	>3600	321	337	−1.23%
	2	325	>3600	321	786	−1.23%
	3	326	>3600	321	513	−1.53%
	4	322	>3600	321	407	−0.31%
	5	325	>3600	320	348	−1.54%
ISG5	6	434	>3600	427	477	−1.61%
	7	433	>3600	427	484	−1.39%
	8	437	>3600	427	561	−2.29%
	9	439	>3600	427	443	−2.73%
	10	449	>3600	427	437	−4.90%
ISG6	11	–	>3600	548	567	–
	12	466	>3600	450	674	−3.43%
	13	548	>3600	548	803	0.00%
	14	562	>3600	548	537	−2.49%
	15	459	>3600	450	750	−1.96%
Average						−1.90%

Notes: (1) F_{CPLEX} and F_{CG} represent the MILP model objective value by CPLEX and column generation-based heuristic algorithm, respectively. And t_{CPLEX} and t_{CG} denote the computation time of CPLEX and algorithm to solve the MILP model, respectively. F_{CPLEX} corresponding to $t_{CPLEX} > 3600$ in the table is the MILP model objective value by CPLEX for $t_{CPLEX} = 3600$. (2) $\Delta_{F_{CG}} = (F_{CG} - F_{CPLEX})/F_{CPLEX}$.

increasing the number of nodes increases the complexity of the delivery process, and a larger number of nodes will increase the computation time. Therefore, decision makes should take these factors into account and should not use too many Decision makers should take these factors into account and should not use too many nodes in the road network.

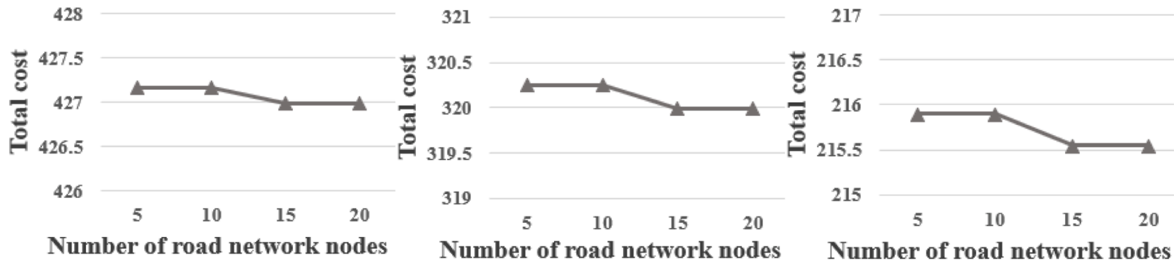
6.3.3. The impact of the speed of drone

Drone speed is a very important parameter, which can affect the solution cost. The drone speed was set as 48 km/h for the previous experiments. In the sensitivity analysis, the drone speed is set as 30, 40, 50, and 60 km/h. We therefore conduct some sensitivity analyses to investigate the effect of this parameter. The results shown in Fig. 5 indicate that when the speed is less than some threshold value, faster drones help reduce the solution cost, but beyond the threshold, faster drones increase the cost. We believe that there are two main reasons for this phenomenon: 1) Faster drones improve the service efficiency of drones, but also affect the travel cost per time unit, which also increases the travel cost imperceptibly. 2) Due to the customer's time window requirements, faster drone speed may not always yield a better performance. The travel routes of drones will be changed as the speed of the drone changes, and the related cost may increase or decrease.



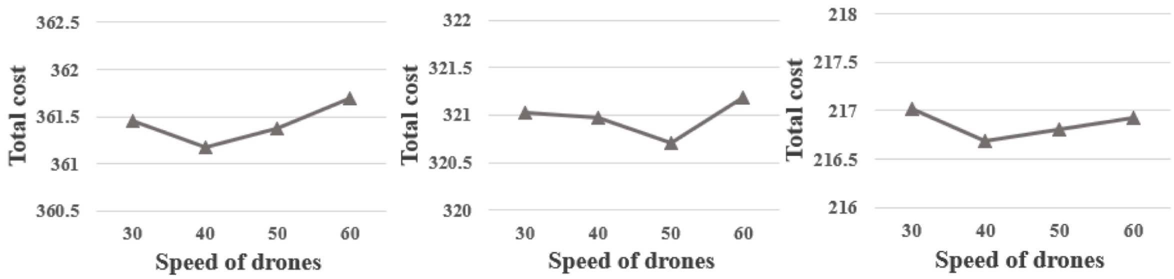
(a) Sensitivity analysis under ISG5 (b) Sensitivity analysis under ISG4 (c) Sensitivity analysis under ISG3

Fig. 3. The impact of the number of drones.



(a) Sensitivity analysis under ISG5 (b) Sensitivity analysis under ISG4 (c) Sensitivity analysis under ISG3

Fig. 4. The impact of the number of nodes in the road network.



(a) Sensitivity analysis under ISG5 (b) Sensitivity analysis under ISG4 (c) Sensitivity analysis under ISG3

Fig. 5. The impact of the speed of drone.

6.3.4. *The impact of the drone maximum flying time*

The model in Section 4 of this paper does not consider the maximum flying time of the drone. This section analyzes whether taking into account the maximum flying time of the drone has an impact on the final result. In this section, we set the maximum flying time of the drone considered in the experiment to 0.5 h. The experimental results are shown in Fig. 6.

As can be seen from the results of Fig. 6, there is no significant difference between the impact of considering maximum flying time and not considering maximum flying time on cost. The cost of considering the drone maximum flying time will be slightly higher than the cost of not considering the drone maximum flying time. This is mainly because the drone has the maximum flying time requirement, which affects the route of the drone. The drone cannot wait for the arrival of the truck without limitation at the nearest road network node, so the UAV needs to choose other routes that meet the drone maximum flying time, which affects the total cost. Not considering the drone maximum flying time allows the drone to fly without time limits, so the drone is able to cover longer distances and more delivery tasks. This is beneficial for decision makers as it increases their delivery scope and capabilities and meets the needs of more potential customers.

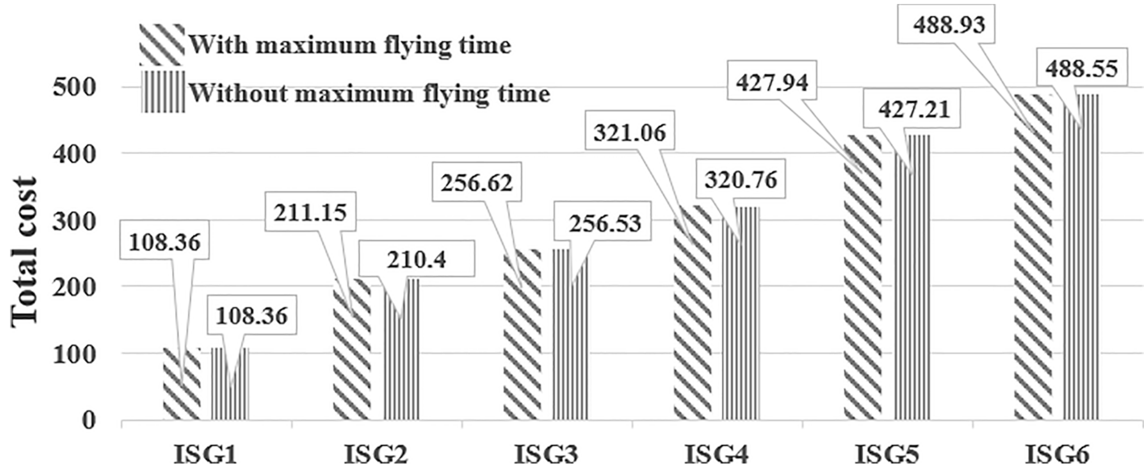


Fig. 6. The impact of the drone maximum flying time.

7. Conclusions

We have described a comprehensive routing and scheduling methodology that can be applied to truck-drone cooperative delivery systems. A mathematical model is established and a column generation-based algorithm for a fleet of truck groups, each of which consists of only one truck and multiple drones. The study's main contribution are the following two aspects.

From modeling perspective, the MILP model is proposed for optimizing truck groups routes, as well as the time at which and locations where drones are launched from and returned to their dedicated trucks. The model was formulated in the context of a road network containing more nodes than in previous studies. Different from most of the related literature that aims to minimize travel time, the proposed model minimizes the total operational cost, including truck travel cost, drone travel cost, using truck groups fixed cost, and potential penalty for late delivery at customer locations. The consideration of many factors makes the proposed model more adaptable to the realistic environment of the truck-drone cooperative delivery.

From the algorithm designing perspective, we have developed a column generation-based algorithm for solving the complex model. While most of the related literature about the TSP-D or the VRP-D adopts the VNS based algorithmic framework, this study may be the first one to incorporate column generation in a heuristic for a VRP-D variant; several acceleration techniques were also designed to decease the algorithm computation time. A VNS based algorithm was also implemented and applied to realistic instances. As to validate the proposed algorithm performance, extensive numerical experiments based on a company's operation network of drones in Hangzhou city were conducted. In addition, the proposed algorithm can work in a rolling horizon manner and solve a batch of orders containing about 30 orders every 15 min. In a realistic context, in which orders emerge and enter the system at an average rate of 120 orders per hour, our proposed methodology can handle the accumulated orders every 15 min.

This study contains some limitations. The algorithm can also be further improved to solve the problem of truck-drone collaborative delivery on a larger scale. The algorithm proposed in this study is a heuristic in nature although the optimality gap is zero in most of experimental instances. In the future, we should try to design some exact solution algorithm for the problem. Last but not the least, it would be an interesting direction to consider both the delivery and the pickup requirements of customers in this truck-drone cooperative delivery mode. All the above could be future research opportunities in the related fields.

CRediT authorship contribution statement

Jiajing Gao: Investigation, Methodology, Data curation, Software, Writing – review & editing, Writing – review & editing. **Lu Zhen:** Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Gilbert Laporte:** Conceptualization, Writing – review & editing, Supervision, Project administration. **Xueting He:** Conceptualization, Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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